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Prediction of residential real estate price on primary market using machine learning

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Abstract

Currently, primary residential real estate market is steadily increasing its prices regardless of the pandemic and national quarantine. There is great potential for financial gain and knowledge in the analysis of this market. Individual analysis seems difficult and time-consuming. This leads to the problem of estimating a favorable price for an apartment. Another example of the use of housing price prediction algorithms can be, for example, an agency engaged in finding, buying and selling properties. In order to maximize profits, the company can invest in an application that can estimate housing prices very accurately. This will reduce the likelihood of overpaying for an investment and give a sticking point for negotiations. In addition, a good price prediction will give you an edge in the real estate trade market. Knowing the value of the house will allow you to lower the price during purchase talks and then sell the apartment at the actual or higher price. The main aim of the paper is to develop a model and application for prediction of residential real estate price in Wroclaw primary market. Data Scraping approach, Simple Linear Regression, Gradient Boosting Regression, Multiple Linear Regression, LASSO, RandomForest Regression methods will be used for this research. In summary, the developed models and application for prediction of residential real estate price in Wroclaw primary market allows for achieving precision about 90%.

Developed application can be used by companies that are considering starting this type of project and, for this purpose, are looking for an example of the entire process, starting with an idea and ending with a statistical model that predicts housing prices based on selected characteristics. The scraping application can be used for commercial as well as private purposes during market analysis. In addition, it can also serve as a basis for new solutions and ventures.

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1. Introduction

Currently, the primary residential real estate market is steadily increasing its prices regardless of the pandemic and national quarantine. There is great potential for financial gain and knowledge in the analysis of this market. One of the reasons for the research may be the desire of an individual to purchase a new apartment in Wroclaw. Due to the lack of advanced tools and techniques, individual analysis seems difficult and time-consuming. This leads to the problem of estimating a favorable price for an apartment. Another example of the use of housing price prediction algorithms can be, for example, an agency engaged in finding, buying and selling properties. In order to maximize profits, the company can invest in an application that can estimate housing prices very accurately. This will reduce the likelihood of overpaying for an investment and give a sticking point for negotiations. In addition, a good price prediction will give you an edge in the real estate trade market. Knowing the value of the house will allow you to lower the price during purchase talks and then sell the apartment at the actual price or higher.

The main aim of the paper is to develop a model and application for prediction of residential real estate price in Wroclaw primary market. For this, it is necessary to select data sources. For example, there are open repositories on the Internet that provide huge amounts of data sets from various fields for free, such as Kaggle, WHO, World Bank Open Data, etc. Due to the low level of detail and inadequacy of the data provided, a Data Scraping approach will be used for this research. This is based on the use of automation scripts that retrieve and record information from selected websites to create a complete data set. Potential sites to use the bot include: Otodom.pl, Domy.pl, Nieruchomości-Online.pl.

The article structure is as follows. Section 2 presents the related works. Section 3 indicates materials and methods, including data source and processing, modeling, and measures. In section 4, the results are presented and discussed. Last sections presents the conclusion, summarizes the paper's findings and indicates the future research directions.

2. Related works

In recent years, the application of machine learning techniques in predicting residential real estate prices has gained significant attention due to its potential to enhance decision-making processes in the real estate industry. This section reviews several relevant studies that have explored similar research objectives and methodologies.

Historically, real estate price prediction primarily relied on conventional statistical methods and econometric models [1],[2]. These approaches often involved the analysis of historical sales data, demographic factors, economic indicators, and local market trends. While these methods provided valuable insights, their predictive accuracy was limited by the complexity and non-linearity inherent in real estate markets [3].

With the advent of machine learning, researchers have sought to leverage its capabilities to overcome the limitations of traditional approaches. Various machine learning algorithms, including regression models, decision trees, support vector machines, and neural networks, have been applied to predict residential real estate prices.

Several studies have employed regression-based algorithms such as linear regression, ridge regression, and lasso regression to model the relationship between real estate attributes (e.g., location, size, amenities) and prices. For instance[4], [5] utilized multiple linear regression to predict housing prices based on features such as square footage, number of bedrooms and bathrooms, and neighborhood characteristics.

Decision trees and ensemble techniques like random forests and gradient boosting have also been widely adopted in real estate price prediction. These methods offer flexibility in handling non-linear relationships and feature interactions. Papers [6],[7] demonstrated the efficacy of random forests in forecasting residential property prices by incorporating spatial variables and neighborhood attributes.

Deep learning models, particularly neural networks have shown promise in capturing intricate patterns in real estate data. The fully connected, LSTM, convolutional [9],[10],[11] and hybrid methods has been used [8].

However existing approaches uses mainly datasets from different static databases. Such approaches do not allow data acquiring near real time. Therefore the main contribution of this research is using Data Scraping approach for data acquiring.

3. Materials and methods

The dataset was extracted from listings on the otodom.pl website and contains over 15,000 records and the following attributes:

1. Offer id - a short string of characters unique to individual cases.
2. District - the location of the apartment from a particular offer.
3. Price of the apartment.
4. Price per square meter of the apartment.
5. Floor on which the apartment is located.
6. Number of floors in the building.
7. Year of construction of the building.
8. Monthly rent.
9. Market - whether the apartment already had an owner.
10. Type of building.
11. State of finishing.
12. Area of the apartment.
13. Number of rooms in the apartment.

The attributes indicated above have various values, mostly textual and numeric. In simple terms, the main concept is based on finding the values of these attributes in the source code of the website and saving them to the appropriate data format in the form of a CSV file. Another important component, without which you can't get started, is WebDriver. WebDriver is an open source tool for automated testing of web applications on multiple browsers. It allows navigation to websites, user input, JavaScript execution and more. ChromeDriver is a standalone server that implements the W3C WebDriver standard. The next element is to search the site for all the necessary attributes and tags. When searching and retrieving data from web sources, you can use Selenium libraries (a web application testing tool that can be successfully used to automate data retrieval), as well as bs4 and beautifulSoup (allows you to quickly and easily search for elements in the page source). The logical concept and the way web crawler's work, is shown in Figure 1. the program starts by launching the browser. It can perform this task with the instance actually fired up or do it in the background depending on the user's preference. However, it is worth mentioning that working in the background saves time by removing the graphical interface and downloading only text files. At the stage of opening a new tab, when the information is not correctly reproduced, the previous data is deleted and the next sales listing is passed. This happens when the page title does not equal “Technical break”. In the second case, if the condition is met, then the program refreshes the page until it loads correctly. Most often, one reload is required for this. All data is first written to variables of the list type, then a DataFrame object is created based on it, which is made possible by the Pandas library. In the last stage, it is saved to a CSV file, but this format is not a necessity. It is also possible to use, for example, xlsx format.

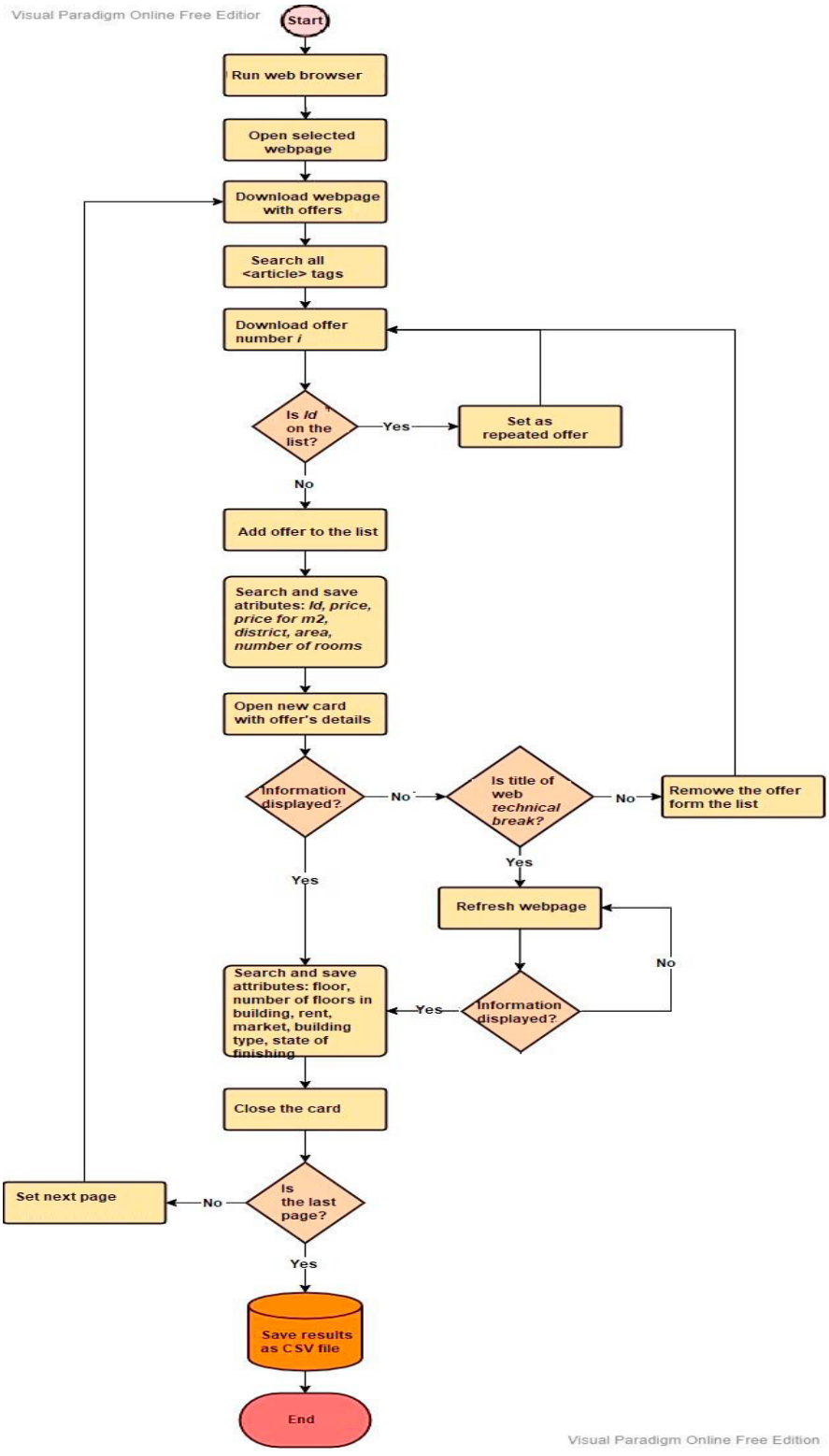


Figure 1. Logical diagram of the web crawler's operation

For predictions based on linear correlation, the predictor (X) and regressor (y) must be selected. In simple terms, this involves selecting the dependent variable and the independent variable. In the case of this experiment, the columns selected are flat_surface_m2 as X and apartment price (price) as y. The next step is to divide the entire data set into two parts. One will be responsible for training the algorithm on it, while the other is to check the accuracy with which the chosen method selects and predicts values. In practice, the size of the test set depends on your needs and mcapabilities, but mostly it is about 20% of the total data. Cross-validation has been used. Once the preparation is complete, one can move on to testing the selected methods. In the case of this research, the following methods were chosen:

1. Simple Linear Regression - is probably one of the most important and widely used regression techniques. It is one of the simplest regression methods. One of its main advantages is the ease of interpretation of the results.
2. Gradient Boosting Regression - Gradient boosting is one of the most powerful techniques for building predictive models.
3. Multiple Linear Regression - is a statistical technique that uses several explanatory variables to predict the outcome of the explanatory variable. The purpose of multiple linear regression (MLR) is to model the linear relationship between the explanatory (independent) variables and the explanatory (dependent) variable.
4. LASSO - is a regression method that performs both variable selection and regularization to increase the predictive accuracy and interpretability of the resulting statistical model
5. RandomForest Regression - is a supervised learning algorithm that uses the ensemble learning method for regression. The ensemble learning method is a technique that combines predictions from multiple machine learning algorithms to make more accurate predictions than a single model.

4. Results

All the results and evaluation of the previously mentioned methods are presented in the table 1:

Table 1. Results of developed methods

Method	**R^2**	**MSE**
Simple Linear Regression	0.607	101 789.11 zł
Gradient Boosting Regression	0.989	5 173.12 zł
Multiple Linear Regression	0.912	41 404.69 zł
LASSO	0.912	41 405. 14 zł
RandomForest Regression	0.986	5 493.95 zł

Evaluation of the performance of the algorithms was presented by three values: R^2 and mean square error. As the table shows, the best results are characterized by two methods: Gradient Boosting Regression (0.989) and RandomForest Regression (0.986).

In the next step, an application was developed that can be used by individual investors and real estate investment institutions. Two best models Gradient Boosting Regression and RandomForest Regression were used. As a last resort, a web application was made that returns the predicted price after entering the appropriate parameters. The application was created on a Web Framework designed for Python called Flask. The first step was to save the trained model into a file. The pickle object storage format was used for this purpose. The pickle module implements binary protocols for serializing and deserializing the structure of Python objects. “Pickling” is the process by which a hierarchy of Python objects is converted into a stream of bytes, while “deserialization” is the reverse operation.

The __pycache__ folder contains files where the byte code of the imported module is cached. The idea is to speed up the loading of python modules by avoiding recompilation (compile once, run many times) when they are re-imported.

They are generated automatically. The selected model is gradient boosting regression. The main file is app.py, which starts the server, opens and loads the model back into a form that can be used for prediction, and has a method that captures the requests sent. Another file contains the data that is passed in the request to the application. Request_flats.py handles connecting to the server and retrieving the housing price prediction from it. To enable the server, run the app.py file on the command line. The next step is to enter specific apartment data in the data_input.py file. Finally, you need to send a request with the offer information so that the result is returned in JSON format (Figure 2):

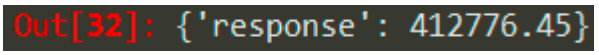


Figure 2. Prediction of real estate price

The user should provide following input data: number of rooms, number of floors, floor on which the apartment is located, year of construction, area in square meters, neighborhood and building type. The output data is the price of apartment.

Conclusions

In summary, the developed models and application for prediction of residential real estate price in Wroclaw primary market allows for achieving precision about 90%.

Developed application can be used by companies that are considering starting this type of project and, for this purpose, are looking for an example of the entire process, starting with an idea and ending with a statistical model that predicts housing prices based on selected characteristics. The scraping application can be used for commercial as well as private purposes during market analysis. In addition, it can also serve as a basis for new solutions and ventures. The main limitation of research is not using deep neural networks for prediction of residential real estate price. It should be main direction of further research. In further research work, to improve the prediction result, an increase in the number of attributes and the data itself can be made. Other sources with free or paid access can also be used. Another thing is to use other imputation possibilities that are not presented in this work.

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